

FARAFRA, Egypt

WMO: 624230

Lat: 27.0583N

Lon: 27.9892E

Elev: 92

StdP: 100.22

Time Zone: 2.00 (E02)

Period: 03-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.2	4.6	-9.0	1.8	20.6	-7.1	2.1	18.5	9.5	17.0	8.4	16.6	1.3	210	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	41.4	20.2	40.1	20.1	39.0	19.8	22.6	35.9	21.8	35.6	21.2	35.4	4.2	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.5	28.6	17.0	12.3	28.5	15.9	11.4	28.5	66.9	35.7	64.0	35.5	61.8	35.9	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.5	7.5	6.7	DB	1.2	44.1	1.1	2.0	0.4	45.5	-0.3	46.7	-0.9	47.9	-1.6	49.4	(4)
			WB	-2.0	23.7	1.6	1.1	-3.2	24.5	-4.1	25.1	-5.0	25.7	-6.2	26.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.8	12.6	15.0	18.7	23.1	26.6	29.8	30.8	31.1	28.5	24.4	18.5	13.9
	DBStd	7.07	2.19	3.45	3.54	3.40	3.21	2.28	1.66	1.62	2.20	2.73	2.76	2.24
	HDD10.0	4	3	1	0	0	0	0	0	0	0	0	0	1
	HDD18.3	489	179	104	36	1	0	0	0	0	0	0	30	140
	CDD10.0	4673	83	141	269	393	515	595	646	654	554	447	254	121
	CDD18.3	2116	1	11	47	144	257	345	388	396	304	189	34	1
	CDH23.3	29478	18	175	666	1819	3489	5125	5946	5988	3913	1961	353	25
CDH26.7	17371	2	59	275	921	2022	3253	3787	3795	2246	912	97	3	
Wind	WSAvg	2.8	2.3	2.5	2.8	3.0	3.1	3.2	3.0	2.8	3.1	2.8	2.4	2.2
Precipitation	PrecAvg	2	0	0	1	0	0	0	0	0	0	0	1	0
	PrecMax	27	1	5	9	0	1	0	0	0	0	4	19	0
	PrecMin	0	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	5	0	1	2	0	0	0	0	0	0	1	4	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	32.6	37.1	39.3	43.1	44.0	42.2	41.8	40.5	38.1	32.4	26.0
		MCWB	12.5	15.5	16.8	17.8	19.9	21.6	20.5	21.4	19.9	19.0	16.5	14.6
	2%	DB	22.6	28.1	33.0	36.8	39.7	41.4	40.4	40.4	38.8	35.2	29.2	23.2
		MCWB	12.0	14.0	15.2	16.9	18.5	19.6	20.3	21.1	19.9	18.9	16.4	13.2
	5%	DB	20.7	25.0	29.5	34.6	37.5	39.6	39.3	39.3	37.2	33.1	27.1	21.5
		MCWB	11.9	12.8	14.2	16.1	17.8	19.5	20.0	20.7	19.8	18.8	15.8	13.0
	10%	DB	19.1	22.7	26.8	32.1	35.2	38.0	38.2	38.1	35.5	31.4	25.1	20.2
		MCWB	11.3	12.2	13.4	15.6	17.4	19.3	20.0	20.3	19.5	18.2	15.2	12.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	16.4	18.7	19.7	21.1	22.3	23.1	24.2	22.6	21.7	19.1	15.7
		MCDB	21.7	28.0	32.4	33.8	38.9	38.7	36.2	35.8	33.2	31.1	27.3	23.0
	2%	WB	13.0	14.8	16.4	18.2	19.5	21.2	22.1	22.9	21.6	20.5	17.7	14.3
		MCDB	20.5	25.6	29.3	33.6	36.2	37.5	36.1	36.3	33.7	31.3	26.1	21.4
	5%	WB	12.2	13.7	14.9	17.0	18.6	20.3	21.4	22.1	20.9	19.7	16.7	13.5
		MCDB	19.7	23.1	27.5	31.8	35.0	36.7	35.9	35.9	33.6	30.4	25.3	20.5
	10%	WB	11.5	12.7	13.9	16.0	17.8	19.7	20.8	21.4	20.2	19.0	15.8	12.7
		MCDB	18.8	21.9	25.6	30.5	33.5	35.9	35.6	35.7	33.3	29.8	24.1	19.6
Mean Daily Temperature Range	5% DB	MDBR	13.6	14.4	15.0	15.8	15.9	15.8	14.6	14.2	13.9	13.5	13.2	13.2
		MCDBR	15.9	17.7	19.0	19.5	18.6	17.7	15.7	15.3	15.7	15.3	15.3	14.6
		MCWBR	8.8	8.7	8.6	8.1	7.4	6.2	6.0	5.8	5.9	6.2	7.5	8.4
	5% WB	MCDBR	14.2	16.0	16.5	16.8	16.5	16.1	14.4	13.7	13.5	13.5	13.3	13.3
		MCWBR	8.2	8.4	8.2	7.8	6.7	6.2	5.9	5.7	5.8	6.2	7.0	8.0
Clear-Sky Solar Irradiance	taub	0.393	0.439	0.489	0.576	0.584	0.509	0.487	0.480	0.481	0.461	0.405	0.392	
	taud	2.019	1.877	1.755	1.600	1.606	1.781	1.849	1.874	1.862	1.898	2.017	2.027	
	Ebn at Noon	831	821	804	748	741	795	812	816	801	789	811	810	
	Edh at Noon	149	187	223	268	267	223	208	201	198	180	148	142	
All-Sky Solar Radiation	RadAvg	4.24	5.15	6.32	7.14	7.75	8.14	7.98	7.48	6.69	5.56	4.54	3.96	
	RadStd	0.10	0.22	0.20	0.17	0.16	0.11	0.07	0.08	0.12	0.07	0.09	0.11	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page